

Shreya Komarabattini

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Education

Purdue University

Bachelor of Science in Computer Science

May 2026

Skills

Programming Languages: Python, Java, JavaScript, C

Web Technologies: Tailwind CSS, React.js

Backend & Data: Node.js, Express, MongoDB

Tools & Platforms: Git, Jupyter Notebook, JavaFX, SceneBuilder

Testing: Cypress, Jest

Projects

ProtoPlay - Interactive Engineering Learning Platform (*Next.js, TypeScript, Tailwind CSS, shadcn/ui*)

LINK: <https://protoplay.vercel.app/>

January 2026

- Built an interactive web platform that teaches engineering concepts through gamified simulations, architecture challenges, and algorithm puzzles, emphasizing learning by building rather than memorization.
- Designed and implemented challenge-driven user flows that evaluate user decisions in real time and provide immediate feedback to reinforce problem-solving and systems thinking.

Healthycal - Mindful Eating Web Application (*React, Vite, Node.js, Express, MongoDB, JWT, Cypress, Jest*)

January 2026

LINK: <https://healthy-cal.vercel.app/>

- Built a full-stack web application for tracking meals, nutrition data, and promote awareness of eating habits.
- Developed a RESTful backend using Node.js and Express with MongoDB for user and meal data storage.
- Implemented JWT-based authentication and authorization for secure user access and protected routes.
- Enabled full CRUD functionality for meal logs tied to individual user accounts.
- Wrote end-to-end tests with Cypress and unit/API tests using Jest.

Fort Wayne Crimes, Purdue University

December 2025

LINK: <https://www.kaggle.com/code/shreyakb/fort-wayne-crimes>

- Analyzed 150,000+ crime records using Python to classify incidents as violent or non-violent.
- Identified crime trends and patterns through data aggregation and visualization techniques.
- Collaborated with a graduate-level research team to produce data-driven reports and insights.

Experience

Undergraduate Researcher: Do We Really Know How to Use Graphs Effectively | Dr. Beomjin Kim

Sept - Feb 2025

- Investigated effective visualization of categorical data, analyzing how users interpret bar, line, and stacked bar charts.
- Designed user studies to evaluate visualization clarity and reduce misinterpretation by non-expert audiences.
- Formulated evidence-based guidelines for selecting appropriate chart types to improve data comprehension.
- Presented findings at the 28th Annual Student Research and Creative Endeavors Symposium.

Systems Software Assistant, Purdue University

Aug 2023 - Present

- Re-imaged, configured, and maintained 100+ lab computers, ensuring consistent OS, software installations, and security updates.
- Resolved 1,500+ hardware and software tickets, maintaining 99% lab uptime and minimizing disruptions for students and faculty.
- Performed routine diagnostics, printer maintenance, and troubleshooting of technical equipment.
- Collaborated with IT staff while independently managing lab systems, providing front-line technical support, and planning and promoting lab events.

Teaching Assistant - Senior Capstone Project, Purdue University

2025

- Assisted in evaluating capstone project reports and presentations, leveraging Python scripts to support data processing and review pipelines.

Summer Camp Assistant, Purdue University

2025

- Represented the Computer Science department during a high-school summer camp program.
- Assisted students with technical activities and project development.
- Conducted research on prompt engineering and evaluated student capstone posters using AI-based tools.

Leadership involvement

- Presented research findings at the Love Data 2025 workshop at Purdue University, discussing data visualization practices and effective interpretation of categorical data.